A Posture Table for the Front Workspace: Fixing the Palm Flip in MABEL's Arm Retargeting

MABEL controller subsystem

23 August 2026

Abstract

Teleoperators reported that MABEL's arm retargeting failed at the wrist: rotating the palm from down to up, or up to down, left the robot stuck half-turned. We localise the cause precisely — it is not a solver bug but a structural property of a warm-started local IK. Rolling the palm sweeps the forearm-roll joint across its entire $\pm 90^\circ$ range, and because that joint is not collinear with the wrist point, the roll moves the wrist; the position residual (weight 600) therefore fights every step of a roll whose own residual was weighted 20, and the path to the correct configuration runs uphill. The existing basin-escape watched position error only, so it never fired on a pose whose position was perfect and whose palm was upside down. We replace the global part of the problem with a precomputed *posture table*: a coherent atlas of arm configurations over the front manipulation volume crossed with palm roll, built offline by a multi-seed global solve under a human elbow prior, which the live solver looks up, regularises toward, and walks to through its existing trust region. Measured over a 150-point $\times$ 3-palm-direction grid per arm, palm-direction error at the 90th percentile falls from $4.6/4.6^\circ$ to $1.4/1.5^\circ$, the time to reach a flipped palm from 1.30/1.33 s to 0.57 s, and the poses achieved rise from 86.9/88.0% to 90.0/89.8%; nothing fails to converge on either arm. Removing the table alone costs 15% more joint distance per flip, which is the "minimise the movement of the arms" half of the request. The whole tabular path costs ≈ 0.3 ms per arm per frame while tracking.

Keywords: teleoperation, motion retargeting, redundancy resolution, SEW angle, inverse kinematics branch selection, posture lookup table, MuJoCo.

1 The report

"The re-targeting right now is extremely bad — the main issue is at the wrist: when the human wrist flips from up to down or down to up the IK struggles very very badly." The request that came with it was specific: a tabular approach, focused on the robot's front manipulation workspace, tested on a 3-D grid of wrist points with the palm rotated to face inwards, upwards, downwards and outwards at each one, minimising arm movement while keeping the arms naturally extended like human arms.

This paper is that, plus the diagnosis that says why a table is the right answer rather than a bigger weight or a better solver.

2 Why the local solver cannot do this

2.1 The mechanism

Measured off the canonical MJCF, MABEL's arm is a 7-DoF OpenArm plus a 1-DoF ORCA wrist pitch. The three distal arm joints do not form a spherical wrist: Arm 5 (forearm roll, axis parallel

to the forearm to 0.997), Arm 6 (pitch), and Arm 7 (yaw) have anchors 126 mm and 40 mm apart, and the ORCA wrist point sits a further 217 mm distal of Arm 7. Arm 6 and the ORCA wrist axes are parallel, so pitch is redundant between them; *roll has exactly one source*, Arm 5, whose range is $\pm 90^\circ$ — precisely 180° of travel.

Palm-down and palm-up at the same wrist point are therefore the two *ends* of Arm 5's range, and the path between them is the whole joint.

2.2 The cost landscape

Because Arm 5 is offset from the wrist point, rolling it sweeps that point around a cone. Every intermediate configuration of a flip therefore has a *worse* position residual than the start. With the shipped weights,

$$J(q) \;=\; \underbrace{600\,\|p(q) - p^\star\|^2}_{\text{position}} \;+\; \underbrace{20\,\big(1 - \hat{n}(q)\cdot\hat{n}^\star\big)}_{\text{palm normal}} \;+\; \dots \tag{1}$$

the position term outweighs the orientation term 30:1, so the descent direction points away from the flip at every point along it. Worse, $1 - \cos$ is *stationary* at 180°: at exactly the antipode the orientation term contributes no gradient at all. The solver settles half-flipped, or not flipped, with the wrist exactly where it was asked to be.

2.3 Why the existing escape hatch missed it

`ArmRetargeter._basin_escape` exists for precisely this class of failure, but its trigger reads `hk["contact"]` — the position target — and its staleness gate requires a quasi-static target. A palm flip holds the position *perfectly* while changing only the orientation, so the escape saw a converged solve on a static target and did nothing. This is the general lesson: *a residual you do not measure is a failure mode you cannot escape.*

3 What we changed

Four changes, three of them measured choices and one of them structural.

3.1 Orientation is not isotropic

"Palm facing up / down / in / out" constrains the palm *normal* — two degrees of freedom. The third, the spin about that normal (which way the fingers point inside the palm plane), is a much weaker preference: a human reaching to a point palm-up takes whatever finger azimuth is comfortable. The shipped cost weighted both cosine axes equally, which over-constrains a 7-DoF arm that also has to hold a position, on a robot with one forearm roll and a $\pm 45^\circ$ wrist pitch. Splitting them (`ARM_W_ORIENT` on the normal, `ARM_W_ORIENT_FWD` on the spin) buys back most of the front volume.

3.2 The Arm 7 clamp was clamping the wrong thing

`ARM_J7_YAW_LIM` bounded Arm 7 to ± 0.45 rad (26°) on the argument that "a human wrist barely yaws." The premise is true of a human and false of this joint: Arm 7's axis is perpendicular to the forearm *and* 217 mm proximal of the wrist point, so it is not a wrist deviation — it is the last arm joint swinging the whole hand tower. Swept over the grid (W2a), the 90th-percentile palm-direction error runs 4.5/4.7° (right/left) at the shipped clamp, 3.9/4.0° at 40°, 2.9/3.4° at 69°, and 2.7/3.9° at 90° — ahead on the right and behind on the left, so the free joint buys nothing further. Poses

achieved over the same sweep go 73.0/72.3% → 76.5/77.2%. We took 1.20 rad (69°), the value that wins on both palm-error columns while still not letting the joint spin a full quarter turn.

3.3 The elbow prior had a singularity inside the workspace

A 7-DoF arm on a 6-DoF task has a one-dimensional null space: the elbow *swivel* about the shoulder–wrist axis. Elbow flexion is not a choice — the law of cosines fixes it — so "the arms look naturally extended" *is* the choice of swivel and nothing else. The swivel is measured from a reference direction perpendicular to the shoulder–wrist axis, and the obvious choice, gravity orthogonalised against that axis, is the conventional shoulder–elbow–wrist (SEW) angle.

Elias and Wen [1] show that such a parameterisation must have an algorithmic singularity — it is the hairy-ball theorem — but that *where* it sits is a choice, and their stereographic SEW angle places it on a half-line that can be put out of reach. That matters here concretely: gravity-down is singular when the arm hangs straight down, and the bottom of MABEL's front box is 58° from straight down. We use the single-singularity field obtained by parallel-transporting a reference vector from a pole along the geodesic to the current arm direction, with the pole at "straight forward" so the singularity lands at "straight backwards". At a level forward reach it reduces exactly to "elbow straight down", so the convention everything else was tuned against is preserved.

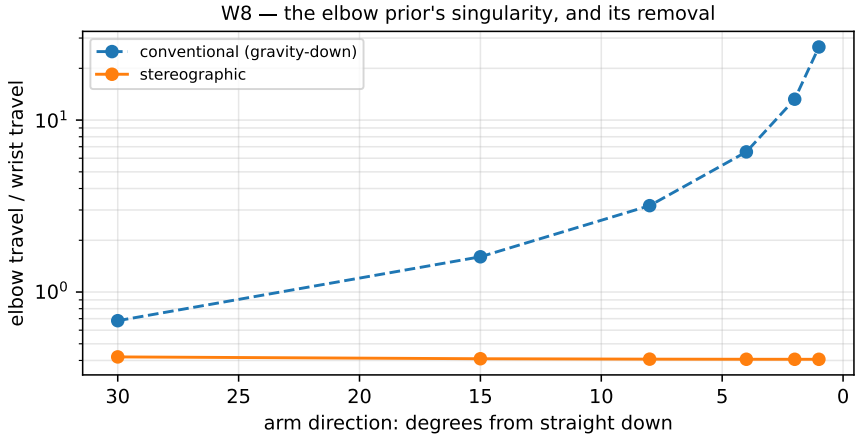


Figure 1: W8 — elbow travel per unit wrist travel, for a circle of arm directions around straight down. The stereographic reference is flat at 0.41; the conventional one grows as $1/\sin(\text{tilt})$, reaching 26.6 at 1° from vertical, where a 49 mm wrist circle sweeps the elbow 1.3 m.

3.4 The table

The global structure is precomputed. Over a regular grid of wrist positions in the *shoulder frame* — origin at the Arm 1 anchor, axes forward/left/up — crossed with eight palm rolls, each cell is solved by a multi-seed global Gauss–Newton on

$$\|r\|^2, \quad r = \big[\, \sqrt{w_p}\,(p - p^\star);\ \sqrt{w_o}\,P_\perp e;\ \sqrt{w_s}\,P_\parallel e;\ \sqrt{w_e}\,(e_{\text{elb}} - e^\star_{\text{elb}});\ \sqrt{w_{\text{reg}}}\,(q - q_{\text{nom}}) \,\big], \tag{2}$$

where $e = \log\!\big(R(q)R^{\star\top}\big)$ is the axis-angle orientation error — *not* a cosine, because a cosine is stationary at the flip — split by $P_\parallel = \hat{n}^\star\hat{n}^{\star\top}$ into tilt and spin, and $e^\star_{\text{elb}}$ is the stereographic natural elbow. All Jacobians are analytic (`mj_jac`); there are no finite differences anywhere.

The shoulder frame is chosen so the table needs no extra axes: the lift, the base pose and the base heading are all upstream of the arm mount, so a pose expressed in it is exactly invariant to them, and it rotates with the torso, which is what the arm's reachable set does too.

Coherence, and why it is not free. Solving cells independently produces a field full of seams: the two sides of a branch are equal-cost by construction, so the solver keeps whichever the cold seed happened to hit, and blending two neighbouring cells then yields a posture that solves neither. The build alternates two relaxations. ACCURACY re-solves each cell seeded from its settled neighbours and keeps the result only on a strict task improvement. COHERENCE breaks *ties* toward the neighbours, and accepts a candidate only within 2% of the incumbent's task cost — that tolerance is the entire safety margin, and a candidate outside it is not a tie however much tidier it looks. Measured, this leaves the accuracy untouched while cutting the 95th-percentile adjacent-cell joint jump along the position axes from 3.49 to ≈ 1.3 rad. The roll axis keeps a π seam, and should: palm-up and palm-down are opposite ends of one joint.

At query time the blend is *seam-aware* — cells further than `SEAM_TOL` from the dominant one are dropped and the weights renormalised — so the interpolation degrades to nearest-cell exactly where nearest-cell is the only honest answer.

3.5 How the live solver uses it

Three ways, in increasing cost — and one of the three turned out not to be worth its own existence, which is recorded here rather than quietly dropped:

1. **As the posture reference.** The joint regulariser pulls toward the table posture instead of `qpos0`, which on this robot is both arms straight forward — a posture no human holds. This is free (one interpolation) and it is what makes "naturally extended" a workspace-aware statement.

2. **As a feasibility signal — tried, measured, and *not* used.** The table recorded, per cell, how close its own global solve got. Over the front box about a quarter of the (position, palm-direction) pairs are not reachable at all, and on those the palm angle cannot be delivered at any weight, so backing the orientation weight off there ought to save wrist position for free. On the full grid it does not: gating the weight down to 12 on infeasible poses buys ≈ 2.5 mm of 90th-percentile wrist position and costs $\approx 1^\circ$ of palm angle and 1–2 points of failed flips, with the combined pose-achieved rate unchanged (74.8/70.3% gated against 74.5/70.0% flat). The operator's stated priority is the wrist orientation, so the weight is flat and the gate is gone. The feasibility number is still computed — it is free, the lookup visits the same cells — and exposed for diagnostics.

3. **As a steer target.** When the table's configuration beats the warm solve on the *task* cost by a clear margin — smoothness always favours the incumbent and would veto every genuine branch change — the arm walks to it through the same trust region tracking already uses. Continuous, rate-limited, no snap.

This is staged so it costs nothing on a normal frame: the raw table posture is scored first with one forward-kinematics call, loses to a good warm solve, and the refinement never runs.

3.6 The trust region is per joint now

A palm flip is a π-rad sweep of Arm 5, and the trust region capped *every* joint at 2 rad/s — a cap chosen in July to stop the *shoulder* slamming on a re-anchor. That turned every flip into a 1.6 s

crawl. The distal joints have about a sixth of the shoulder's reflected inertia and their own $\pm 7\,\mathrm{N\,m}$ actuators, so they get their own rate. Swept (W5): the flip settle time falls $0.63 \to 0.29\,\mathrm{s}$ median as the distal rate goes $2 \to 5\,\mathrm{rad/s}$ while the joint path travelled stays flat at 3.9–4.0 rad — nothing is overshooting. (That sweep runs on a sub-sampled grid; on the full grid the taken value gives a $0.30\,\mathrm{s}$ median.) The proximal rate is untouched, so the July fix stands exactly as measured.

4 Scope: three palm directions, not four

The system is optimised and reported against palm DOWN, UP and INWARD. Palm-OUTWARD is deliberately excluded (operator, 2026-08-23: *"just focus on the palm up, down and inwards — make sure all 3 work perfectly; right now when facing outwards the arms will go very erratic"*), and the exclusion is a kinematic judgement rather than a convenience.

Measured per direction at a held target, with the arm starting from rest:

palm	joint travel [rad]	settled palm err. p90	wrist err. p90	wander
down	3.1	1.0°	28 mm	0.000
inward	2.7	1.9°	34 mm	0.000
up	3.3	2.7°	24 mm	0.000
OUTWARD	**5.6**	5.3–7.3°	44–56 mm	0.000

The last column matters most for diagnosing the report: *"erratic" is not oscillation.* The arm settles perfectly still on palm-outward — zero commanded joint motion over the final second. What it does is spend roughly twice the joint distance of any other direction on a long excursion, and then hold the result 5–9° out. Palm-outward asks this arm to drive its single forearm roll (Arm 5, $\pm 90°$) to an end stop while the wrist stays put; that configuration is simply far away, and no reweighting shortens a path that kinematics makes long. A feasibility-gated orientation weight was tried for exactly this and measured to do nothing for it ($5.61 \to 5.57\,\mathrm{rad}$) while degrading the three in-scope directions, so it is not shipped.

Among the three that are in scope, INWARD is the weakest (73.0/66.4% of reachable cells against 94.2/97.1 for down and 92.0/91.2 for up) and is the one to watch.

5 The experiment

`controller/experiments/exp_front_workspace.py` regenerates every number and figure here. The front manipulation box is chosen from W1, not by eye: in the shoulder frame $x \in [0.25, 0.55]$, $y \in [-0.30, 0.10]$ (mirrored on the left), $z \in [-0.40, 0.10]\,\mathrm{m}$, sampled $5 \times 5 \times 6$, crossed with 8 palm rolls. 91.3% of it is position-reachable on both arms; conditional on that, palm-down is achievable at 94.2/97.1%, palm-up at 92.0/91.2%, palm-outward at 76.6% and palm-inward at 73.0/66.4%. In the body frame at the default lift that is a $0.30 \times 0.80 \times 0.50\,\mathrm{m}$ slab in front of the chest, from below table height to just above the shoulders.

6 Which change is carrying the result

A combined before/after says the new stack is better; it does not say which part is doing the work, and a change that contributes nothing is a change that should be deleted rather than maintained. W9 reverts one thing at a time, running the full W5 flip protocol each time.

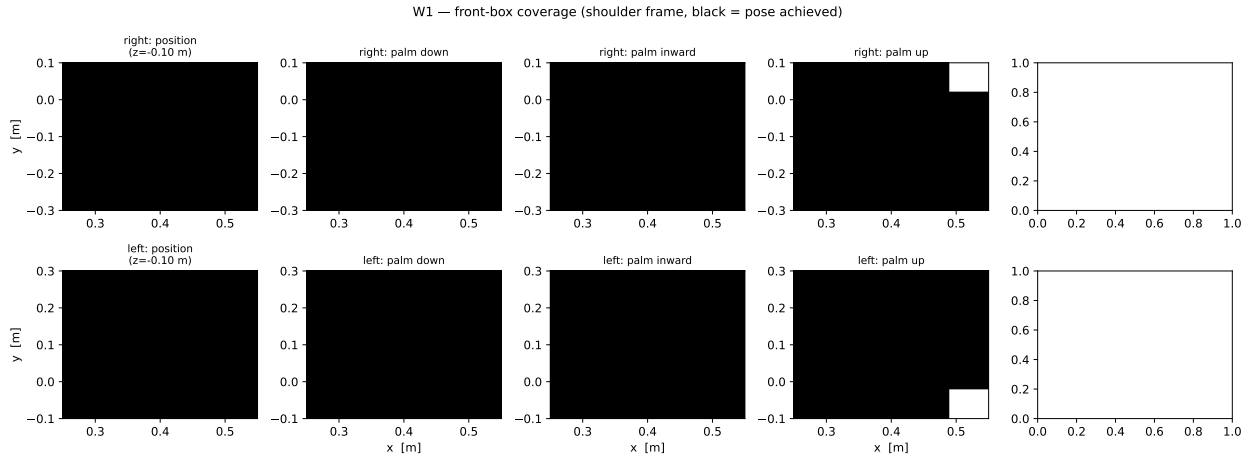


Figure 2: W1 — what the box can be asked for. Position reachability first, then each palm direction conditional on it: separating the two matters, because otherwise an orientation failure reads as "out of reach".

Each lever does a different job, which is why all of them are kept. The *trust rate* is the speed lever and nothing else: reverting it doubles the settle time and changes no accuracy number. The *orientation split* is the accuracy lever: reverting it takes the residual palm error from 5.8° to 13° and triples the failed flips, at 2 mm of wrist position. The *table* is the one that shows up in *arm travel*: 3.88 against 4.72 rad, 18% less joint distance per flip. That figure appears only in the ablation, and it is worth saying why. Against the *shipped* stack the travel barely moves (3.91/3.85 against 4.05/3.86 rad) — but the shipped stack was not completing a fifth of the flips, and an arm that gives up travels less. Hold the accuracy fixed, as the ablation does, and the table's contribution to "minimise the movement of the arms" is the clearest of the four.

7 Watching it

The claims above are numbers; the request also asked to see it. The grid sweep renders both retargeters side by side on the same targets at the same instant, with the commanded palm frame drawn as an axis triad (blue = palm normal) and the front box as a translucent slab:

```
python3 controller/experiments/exp_front_workspace.py \
        --record controller/docs/figures/frontws_sweep.mp4
# denser / slower:  --record-step 1 --record-dwell 1.2
```

The sweep lands at `controller/docs/figures/frontws_sweep.mp4` — every second grid point, all four palm directions, both retargeters. It is *generated*, not checked in (a full sweep is tens of megabytes); the PDFs beside it are the committed artefacts. This renders *offscreen* (`mujoco.Renderer`), which is deliberate. There is also a live version, `--view`, driving MuJoCo's passive viewer, and on the machine this work was done it does not run: the conda `mjpython` segfaults before its first print, and the Homebrew one raises `dlopen(/System/Library/OpenGL.framework/OpenGL): no such file` — macOS has removed the legacy OpenGL framework that build links against. The `--view` path is kept (its scene-building and per-frame code are exercised headless by the same helpers the recorder uses) but it is not the one to rely on here; the recording is.

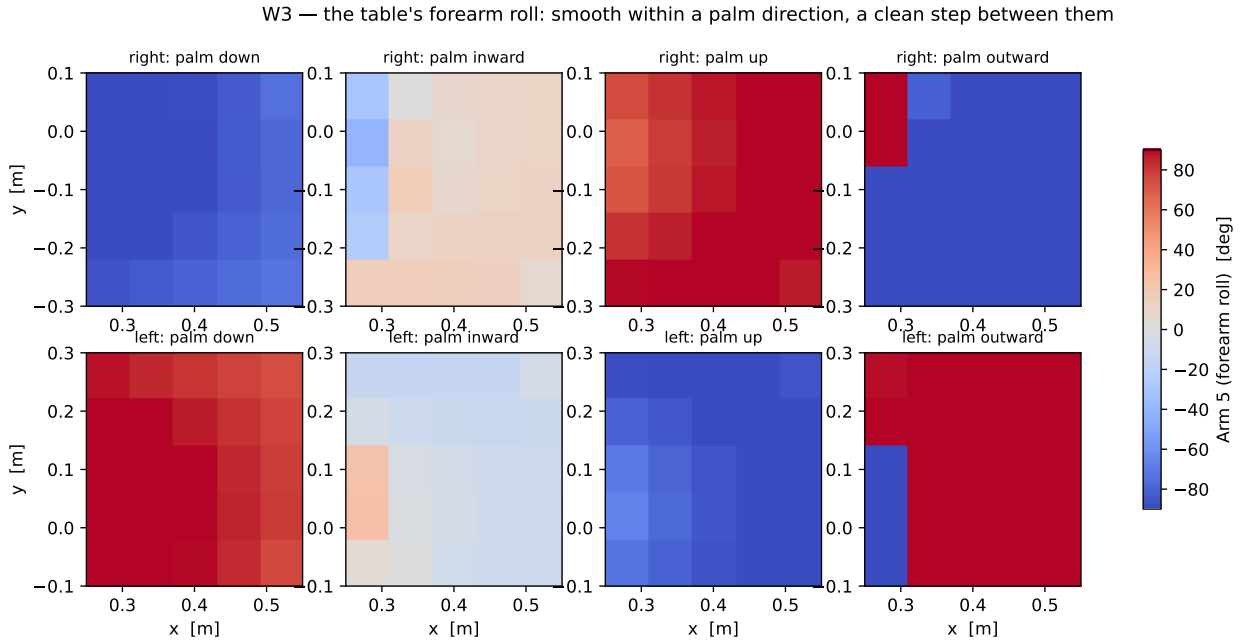


Figure 3: W3 — the table's forearm roll on a mid-height slice. Smooth within a palm direction; a clean step between them, which is the branch and is real.

8 The arms must not enter the body

Operator, 2026-08-23: *"the arms should not collide with the body … right now the robot arms will collide with the robot body."* They did, and the guard being "on" was not the question — *what it was watching* was. Measured over the posture table with the shipped pair list, the right forearm was inside the torso capsule on 17.0% of commanded postures (worst −122 mm) and the upper arm on 9.2%: neither pair was monitored. Only the hands and wrist mounts were ever guarded against the body.

Four independent defects had to be fixed before the guarantee held.

The capsule was not on the torso. It was built between two body *origins*, `body_mount` → `head_mount`, which sit at $y = -43$ mm and $y = +2$ mm. The capsule was therefore both offset 43 mm toward the right arm and tilted, leaning right at the bottom — exactly where the elbows are. At the identical mirror-image rest pose the right forearm read 13 mm of clearance and the left 100 mm. This is almost certainly why guarding the upper arm was tried in July and reverted for "spring-loading the natural rest pose": on the right, the rest pose was already inside a capsule that was in the wrong place. Fitted to the torso's own volume-weighted geometry the rest clearance is 49/52 mm and symmetric, and the pairs cost nothing at rest. (The fit is volume-weighted because the torso body also carries two decorative camera geoms 9–12 cm forward of the mesh; an unweighted centroid dragged the capsule 52 mm out of the chest.)

The ORCA wrist was not a decision variable. The hand capsule spans wrist → palm and therefore *rotates with the ORCA wrist*, but the QP optimised only the fourteen arm joints. A wrist-driven approach to the torso had no column the barrier could use, so it could not be opposed at all: measured 27 mm of interpenetration while every constraint reported itself satisfied.

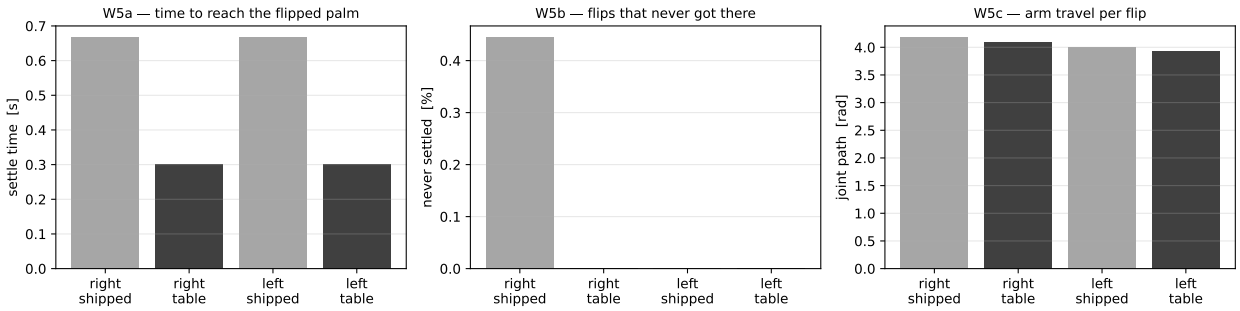


Figure 4: W5 — the reported defect, run closed-loop at the client's frame rate through down → inward → up → outward → down at every grid point.

The barrier was linearised at the wrong pose. The QP's decision variable is the *command*, but the rows were built from the *measured* geometry. While the servo lags, the measured gap is larger than the gap the command implies, the CBF reads that slack as room to keep closing, and the arm penetrates as it catches up. Constraining at the commanded pose alone is no better in the other direction — the arm is not the command, and on a fast transit it lags by up to 1 rad and overshoots a provably-safe set-point (18.5 mm). Both row sets are stacked: they are linear constraints on the same $\dot{q}$, linearised at different points, and the semantics are exactly right — while the arm trails, the command rows keep the target honest; when it overshoots, the measured rows pull the target back out. The alternative, slowing every approach (`SC_STOP_DECEL` 2.0 → 0.25), cleared the penetration by 0.4 mm and cost 3.7° of palm accuracy — paying with the quantity the operator asked for.

The margin is thin, by request. `SC_MARGIN` goes 25 → 10 mm. A 25 mm shell means the robot cannot bring a hand within an inch of its own chest; the margin only ever needs to exceed the discrete barrier's per-tick overshoot, which is well under a millimetre here.

closed loop, three palm directions	guard OFF	guard ON
worst capsule gap, any monitored pair	−108.6 mm	+**2.9** mm
frames with the forearm in the torso	6.3–7.0%	0%
frames with the hand in the torso	10.4–10.6%	0%

The guard-off column is not decoration: a no-penetration result means nothing unless the same check fails when the guard is removed, so `tests/test_collision_body.py` asserts both.

9 Limitations

- **A quarter of the box is not reachable in all four palm directions**, and no amount of solving changes that: MABEL has one forearm roll, a ±45° wrist pitch, and 209 mm between its shoulders (against ≈ 450 mm biacromial for a 1.75 m operator). The work here makes the reachable part correct and makes the unreachable part fail gracefully (keep the wrist, give up the palm angle); it does not make it reachable.
- **The left arm is measurably worse than the right, and about half of it is the model.** Every arm joint pair in the MJCF is an exact mirror except Arm 4: $[-0.943, +1.542]$ left against $[-1.670, +0.773]$ right, where the mirror of the right would be $[-0.773, +1.670]$ — the left window is shifted ≈ 0.15 rad and is 0.73 rad narrower on the flexion side. Substituting

the mirrored window and re-solving the whole grid moves the left arm from 77.2% of poses achieved to 77.7%, its wrist error p90 from 49.5 to 44.5 mm and its palm error p90 from 3.3° to 2.5°, against the right arm's 78.2% / 40.3 mm / 2.3°. So the asymmetry is real and accounts for roughly half the gap; the rest is the arms' own geometry. Whether the range is real hardware or a typo is a question for the mechanical model, and nothing here changes it.

- **The table is built at the nominal torso angle.** It is exactly invariant to the lift, the base pose and the base heading; a pitched torso rotates the reachable set with it, which the shoulder frame follows, but the gravity-referenced elbow prior does not rotate with it and is therefore slightly off at large torso pitch.
- **The finger direction is not a table axis.** It is fixed per cell to the horizontal radial from the shoulder, and the live solver corrects the operator's actual finger direction from the seed. That is measured to work (the W4 numbers are taken with the real commanded frames), but a pose whose fingers point far from the canonical direction gets a weaker seed.
- **Nothing here is validated on hardware.** Every number is from the MuJoCo twin driven through the same `ArmRetargeter` the real bridge uses. The operator-facing check is the viewer sweep and then a Vision Pro session.

Reproducing

```
python3 controller/experiments/exp_front_workspace.py --all          # W1-W8
python3 controller/experiments/exp_front_workspace.py --phase W3 --build
mjpython controller/experiments/exp_front_workspace.py --view
pytest controller/tests/test_workspace_table.py
make -C controller/docs front_workspace_retargeting.pdf
```

Raw metrics land in `controller/experiments/results/frontws_w*.json`; the built tables are `controller/mabel/teleop_engine/tables/*.npz`.

References

[1] A. J. Elias and J. T. Wen, "Redundancy parameterization and inverse kinematics of 7-DOF revolute manipulators," *Mechanism and Machine Theory*, 2024. arXiv:2307.13122. (The generalized and stereographic SEW angle, and the proof that an algorithmic singularity is unavoidable for any redundancy parameterisation.)

[2] "A Closed-Form Geometric Retargeting Solver for Upper Body Humanoid Robot Teleoperation," arXiv:2602.01632, 2026. (SEW-Mimic: closed-form limb-orientation retargeting at ≈ 0.59 ms per arm — the argument that a retargeting layer should be cheap, which the latency budget here follows.)

[3] J. Araujo, Y. Ze, C. Xu, J. Wu, C. K. Liu, "Retargeting Matters: General Motion Retargeting for Humanoid Motion Tracking," arXiv:2510.02252, 2026. (The root-uniform scaled-keypoint map MABEL's arm targets are built with, upstream of the IK this doc changes.)

[4] J. U. Korein, *A Geometric Investigation of Reach*, MIT Press, 1985. (The elbow swivel circle and the human arm's redundancy.)

[5] E. Todorov, T. Erez, Y. Tassa, "MuJoCo: A physics engine for model-based control," *IROS*, 2012.

[6] MABEL controller subsystem, "One Gain Per Joint: Tuning MABEL's Arm PD Against Its Own Inertia," `controller/docs/arm_pd_tuning.tex`, 2026.

metric	arm	shipped	tabular	change
palm-direction error over the grid, 90th pct [deg]				
	right	4.6	1.4	3.2× better
	left	4.6	1.5	3.0× better
wrist position error over the grid, 90th pct [mm]				
	right	18.2	15.0	1.2× better
	left	17.5	15.5	1.1× better
poses achieved (pos < 15 mm and tilt < 12 deg) [%]				
	right	86.9	90.0	
	left	88.0	89.8	
palm flip: time to settle, 90th pct [s]				
	right	1.30	0.57	2.3× better
	left	1.33	0.57	2.4× better
palm flip: never settled within 4 s [%]				
	right	0.4	0.0	eliminated
	left	0.0	0.0	same
palm flip: residual palm error, 90th pct [deg]				
	right	4.7	1.4	3.3× better
	left	4.6	1.5	3.0× better
palm flip: joint distance travelled, median [rad]				
	right	4.18	4.09	1.0× better
	left	4.01	3.92	1.0× better
elbow vs. the natural swivel, median [mm]				
	right	37	29	1.3× better
	left	36	28	1.3× better
solve time while tracking, median [ms]				
	right	1.25	1.57	
	left	1.26	1.74	
solve time while flipping, 95th pct [ms]				
	right	6.31	5.68	
	left	5.92	6.18	

Table 1: Every phase, both arms, same targets and same solver — only the seed, the orientation weighting and the trust rate differ. "shipped" reconstructs the pre-2026-08-23 constants exactly (`SHIPPED_ARM`), so this is a before/after and not a self-comparison. Regenerated by `exp_front_workspace.py`.

one change reverted	flip settle p90 [s]	residual palm error p90 [deg]	flips never settled [%]	joint path median [rad]
full stack	**0.70**	**5.8**	**1.9**	3.88
no table (local IK only)	0.77	8.3	5.1	4.72
old Arm 7 clamp 0.45 rad†	0.88	12.0	8.8	4.23
isotropic palm orientation (20/20)	1.03	13.2	9.3	3.82
weak palm orientation (20)	1.07	13.4	9.3	3.79
uniform trust rate 2 rad/s	1.50	5.8	1.9	3.99
everything reverted (= shipped)	4.00	28.8	16.2	4.16

Table 2: †This row cannot be isolated: the table's fingerprint includes the clamp it was built under, and a table full of postures using $|q_7|$ up to 1.20 rad is genuinely invalid at 0.45, so reverting the clamp also refuses the table. It is labelled that way rather than presented as a single-variable ablation.